Raghavendra Kulkarni

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Education

Northeastern University — Boston, MA
Master of Science in Computer Science (GPA: 3.86) — *Aug. 2024 – Dec. 2026*

Visvesvaraya Technological University — Bengaluru, India
Bachelor's in Information Science and Engineering — *Aug. 2018 – July 2022*

Experience

Software Engineer Intern — Jan. 2026 – Aug. 2026
Vicor Corporation — *Andover, MA*

- Built and scaled **Java/Spring Boot REST APIs** integrating SAP ERP and PLM data into a distributed engineering-data platform, processing **10,000+** daily transactions through batched requests and server-side caching.
- Diagnosed a data-loading bottleneck slowing a PCB workflow, and cut load time from **80s to 2s** for **200+** engineers by introducing lazy loading, request consolidation, and caching.
- Architected **CI/CD** pipelines using **GitHub Actions** and **Docker**, cutting deployment time by **45%** and accelerating release cycles.
- Refactored tightly coupled service and DAO components for testability and built **JUnit 5/Mockito** suites covering controllers, services, and SAP integrations, achieving **95%+** coverage and reducing regression defects by **50%**.

AI Software Engineer (Research Assistant) — Sep. 2025 – Dec. 2025; Sep. 2026 – Present
DASH AI Hub, Northeastern University — *Boston, MA*

- Led end-to-end development of an AI essay-grading platform with **Next.js, Flask, and LangGraph**, reducing manual grading time from **8+ hours to under 30 minutes**, adopted by 10+ faculty across 4 courses serving 300+ students.
- Built a **LlamaIndex/Qdrant RAG pipeline** that retrieves course-specific materials using metadata filters, then tested answer relevance and accuracy with faculty across representative student questions.
- Architected a presentation-coaching platform with **Next.js, FastAPI, and PostgreSQ**L, tracking practice sessions and grading delivery across a multi-agent, 12-category rubric with personalized voice clones generated using F5-TTS.
- Built a distributed GPU pipeline with **Celery** workers, **RabbitMQ** job queues, and **Redis** progress tracking for Whisper transcription, LLM grading, and voice cloning, keeping grading responsive during long-running synthesis jobs.

Software Engineer — Oct. 2022 – Aug. 2024
Cognizant Technology Solutions — *Bengaluru, India*

- Contributed **React/Python** components to a multi-cloud provisioning platform enabling infrastructure lifecycle management across AWS, Azure, and SAP HANA Cloud, replacing manual, SSH-based server operations.
- Wrote **Terraform** based IaC modules and **Kubernetes** manifests/Helm charts to automate infrastructure provisioning and manage scaling and configuration across distributed cloud environments.
- Built Python automation for provisioning and lifecycle workflows across cloud environments, cutting manual setup by **40+** hours per environment across **500+** enterprise tenants.

Software Engineer Intern — *Apr. 2022 – Sep. 2022*
Cognizant Technology Solutions — *Bengaluru, India*

- Migrated core modules from **.NET to Java and Angular** and optimized **SQL** queries in a high-traffic distributed system, cutting page latency by **40%** and eliminating recurring timeouts for **300K+** users.

Projects

Learnbot | *React, Flask, Pinecone, Claude, RAG*

- Designed & built a React/Flask RAG teaching assistant used by 50+ finance students, combining Canvas content ingestion, Pinecone semantic search, and Claude to deliver course-specific guidance within academic-integrity constraints.

DASH API | *React, Flask, REST APIs, Linux, GPU Infrastructure*

- Architected a self-service platform for accessing open-source LLM models on an on-premises GPU server, enabling faculty and students to reserve time slots and receive time-bound access through REST APIs or SSH.

PlanVoyage | *React, Vite, Node.js, Express, Prisma, PostgreSQL*

- Built a trip-planning application that lets users discover destinations and nearby attractions, create and manage itineraries, and share travel plans, with JWT-based authentication and persistent storage through Prisma and PostgreSQL.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
Frameworks: Spring Boot, React, Next.js, Angular, Flask, FastAPI, Node.js, JPA/Hibernate, JUnit, Mockito
AI/ML & Databases: NLP, LangGraph, LlamaIndex, LangChain, Qdrant, Pinecone, PostgreSQL, MongoDB, Oracle, Redis
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Terraform, RabbitMQ, Jenkins, GitHub Actions, Linux, Git
Core Competencies: Data Structures & Algorithms, System Design, Microservices, Distributed Systems, Version Control, SDLC, Agile
AI Tools: Claude Code, Codex, Github Copilot, Open-source LLMs, MCP